Homework 7

MAT 2040 : Summer 2023 : Prof. Lun Yu

Problem 1

Find the eigenvalues and the corresponding eigenspaces for each of the following matrices:

1. $\begin{pmatrix} 3 & 2 \\ 4 & 1 \end{pmatrix}$
2. $\begin{pmatrix} 6 & -4 \\ 3 & -1 \end{pmatrix}$
3. $\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$
4. $\begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{pmatrix}$
5. $\begin{pmatrix} 1 & 2 & 1 \\ 0 & 3 & 1 \\ 0 & 5 & -1 \end{pmatrix}$

Answer

For an $n \times n$ matrix A, its eigenvalues are the solutions to the characteristic equation $\det(A - \lambda I) = 0$, and the corresponding eigenspace for eigenvalue λ is the null space of $A - \lambda I$.

1. Eigenvalues are $\lambda_1 = 5, \lambda_2 = -1$. Corresponding eigenspaces are $\text{Span}\left\{(1,1)^T\right\}$ and $\text{Span}\left\{(1,-2)^T\right\}$.
2. Eigenvalues are $\lambda_1 = 2, \lambda_2 = 3$. Corresponding eigenspaces are $\text{Span}\left\{(1,1)^T\right\}$ and $\text{Span}\left\{(4,3)^T\right\}$.
3. Eigenvalue is $\lambda = 0$. Corresponding eigenspace is $\text{Span}\left\{(1,0,0)^T\right\}$.
4. Eigenvalues are $\lambda_1 = 2, \lambda_2 = 1$. Corresponding eigenspaces are $\text{Span}\left\{(1,1,0)^T\right\}$ and $\text{Span}\left\{(1,0,0)^T, (0,1,-1)^T\right\}$.
5. Eigenvalues are $\lambda_1 = -2, \lambda_2 = 1$ and $\lambda_3 = 4$. Corresponding eigenspaces are $\text{Span}\left\{(1,1,-5)^T\right\}$, $\text{Span}\left\{(1,0,0)^T\right\}$ and $\text{Span}\left\{(1,1,1)^T\right\}$.

Problem 2

Let A be an $n \times n$ matrix. Prove that A is singular if and only if $\lambda = 0$ is an eigenvalue of A.

Answer

Proof. A is singular if and only if the homogenous equation $A\mathbf{x} = \mathbf{0}$ has a non-zero solution. If such a solution exist, and let it be $\mathbf{x}_0$, then we have $A\mathbf{x}_0 = 0\mathbf{x}_0$ and $\lambda = 0$ is an eigenvalue of A. Conversely, an eigenvalue 0 implies the equation $A\mathbf{x} = 0\mathbf{x}$ has a non-zero solution, which indicates A is singular.

Problem 3

1. Let A be a nonsingular matrix and let λ be an eigenvalue of A. Show that $1/\lambda$ is an eigenvalue of A^{-1}.

Answer

1. Proof. λ being an eigenvalue of A means the equation $A\mathbf{x} = \lambda\mathbf{x}$ has a non-zero solution for $\mathbf{x}$. Let $\mathbf{x}_0$ be such a solution (an eigenvector of A), let $\mathbf{y}_0 = A\mathbf{x}_0 = \lambda\mathbf{x}0$. Because A is invertible, we have $\mathbf{y}_0 \neq \mathbf{0}$ and $A^{-1}\mathbf{y}_0 = A^{-1}A\mathbf{x}_0 = \mathbf{x}_0 = \frac{1}{\lambda}\mathbf{y}_0$, therefore, $1/\lambda$ is an eigenvalue of A^{-1}.

Problem 4

1. An $n \times n$ matrix A is said to be idempotent if $A^2 = A$. Show that if λ is an eigenvalue of an idempotent matrix, then λ must be either 0 or 1.

Answer

1. Eigenvalues are solutions to the equation $|A-\lambda I| = 0$. For the idempotent matrix A, we can construct the following multiplication and have: $(A-\lambda I)(A+(\lambda-1)I) = A^2-A-\lambda(\lambda-1)I = -\lambda(\lambda-1)I$. Then, we have the determinant $|(A-\lambda I)(A+(\lambda-1)I)| = -\lambda(\lambda-1)$. Now, consider if λ is an eigenvalue of A, we must have $|A-\lambda I| = 0$ and because $-\lambda(\lambda-1) = |(A-\lambda I)(A+(\lambda-1)I)| = |(A-\lambda I)||(A+(\lambda-1)I)| = 0$, λ must be either 0 or 1.

Problem 5

1. Let A be an $n \times n$ matrix and let $B = A + I$. Is it possible for A and B to be similar? Explain.

Answer

1. Because $B = A + I$, the characteristic equation for A is $\det(A - \lambda I) = 0$ and for B is $\det(B-\lambda I) = \det(A-(\lambda-1)I) = 0$. We see that if λ_0 is an eigenvalue of A, then $\lambda_0 + 1$ is an eigenvalue of B. Since A and B have different sets of eigenvalues, they cannot be similar.

Problem 6

1. Let Q be an orthogonal matrix. Show that if λ is an eigenvalue of Q, then $|\lambda| = 1$.

2. Show that $|\det(Q)| = 1$.

Answer

1. If λ is an eigenvalue of Q, and let $\mathbf{x}_0$ be a corresponding eigenvector, then we have $Q\mathbf{x}_0 = \lambda\mathbf{x}_0$ and $\mathbf{x}_0 \neq \mathbf{0}$. Because Q is an orthogonal matrix, we have:

$$\lambda^2\mathbf{x}_0^T\mathbf{x}_0 = (\lambda\mathbf{x}_0)^T(\lambda\mathbf{x}_0) = (Q\mathbf{x}_0)^T(Q\mathbf{x}_0) = \mathbf{x}_0^T\left(Q^TQ\right)\mathbf{x}_0 = \mathbf{x}_0^T\mathbf{x}_0$$

   Because $\mathbf{x}_0^T\mathbf{x}_0 \neq 0$, the above equation implies $\lambda^2 = 1$, or $|\lambda| = 1$.

2. For orthogonal matrix Q, we have $Q^T = Q^{-1}$, and,

$$(\det(Q))^2 = \det(Q)\det\left(Q^T\right) = \det(Q)\det\left(Q^{-1}\right) = \det\left(QQ^{-1}\right) = \det(I) = 1.$$

   Hence, $|\det(Q)| = 1$.

Problem 7

1. In each of the following, factor the matrix A into a product XDX^{-1}, where D is diagonal:

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

2.

$$A = \begin{bmatrix} 2 & 2 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & -1 \end{bmatrix}$$

Answer

1. The eigenvalues of this matrix is $\lambda_1 = 1$ and $\lambda_2 = -1$. And $x_1 = (1,1)^T$ is an eigenvector belonging to eigenvalue $\lambda_1, x_2 = (-1,1)^T$ is an eigenvector belonging to eigenvalue λ_2, so A can be written as

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

2. The calculation process is the same with (a), A can be written as

$$A = \begin{bmatrix} 2 & 2 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 \\ 0 & -1 & -3 \\ 0 & 0 & 3 \end{bmatrix}\begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}\begin{bmatrix} 1 & 2 & \frac{5}{3} \\ 0 & -1 & -1 \\ 0 & 0 & \frac{1}{3} \end{bmatrix}$$

Problem 8

1. For each of the matrices in the previous exercise, use the XDX^{-1} factorization to compute A^6.

Answer

1. $A^6 = XD^6X^{-1}$, so

$$A^6 = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1^6 & 0 \\ 0 & (-1)^6 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

2.

$$A^6 = \begin{bmatrix} 1 & 2 & 1 \\ 0 & -1 & -3 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 2^6 & 0 & 0 \\ 0 & 1^6 & 0 \\ 0 & 0 & (-1)^6 \end{bmatrix} \begin{bmatrix} 1 & 2 & \frac{5}{3} \\ 0 & -1 & -1 \\ 0 & 0 & \frac{1}{3} \end{bmatrix} = \begin{bmatrix} 64 & 125 & 105 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Problem 9

Let A be a diagonalizable matrix whose eigenvalues are all either 1 or -1 . Show that $A^{-1} = A$.

Answer

Assuming $A = XDX^{-1}$ where $D = \operatorname{diag}(d_1, d_2, \ldots, d_n)$ and $d_i = 1/-1$. We get $D^{-1} = D$, So $A^{-1} = \left(XDX^{-1}\right)^{-1} = XD^{-1}X^{-1} = XDX^{-1} = A$.

Problem 10

Show that any 3×3 matrix of the form $\begin{bmatrix} a & 1 & 0 \\ 0 & a & 1 \\ 0 & 0 & b \end{bmatrix}$ is defective.

Answer

Because the eigenvalues are $\lambda_1 = \lambda_2 = a, \lambda_3 = b$. When the eigenvalue is a, $(A - aI)X = 0$, we can get the eigenvector $X_1 = X_2 = (1, 0, 0)^T$. According to the definition of defective matrix, we know that matrix A is defective.

Problem 11

For each of the following, find all possible values of the scalar α that make the matrix defective or show that no such values exist.
(a)

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & \alpha \end{bmatrix}$$

(b)

$$A = \begin{bmatrix} 3\alpha & 1 & 0 \\ 0 & \alpha & 0 \\ 0 & 0 & \alpha \end{bmatrix}$$

Answer

(a) We firstly calculate the eigenvalues of A are $\lambda_1 = 0, \lambda_2 = 2, \lambda_3 = \alpha$.

- When $\alpha = 0$, we can get the eigenvectors are $X_1 = (1, -1, 0)^T, X_2 = (1, 1, 0)^T, X_3 = (0, 0, 1)^T$.
- When $\alpha = 2$, the eigenvectors are $X_1 = (1, -1, 0)^T, X_2 = (1, 1, 0)^T, X_3 = (0, 1, 1)^T$.

So no such α exist.
(b) The eigenvalues of A are $\lambda_1 = \alpha, \lambda_2 = \alpha, \lambda_3 = 3\alpha$.

- When $\alpha = 0$, the eigenvectors are $X_1 = (1, 0, 0)^T$ and $X_2 = (0, 0, 1)^T$. There are 2 linearly independent eigenvectors corresponding to eigenvalues $\lambda_{1,2,3} = 0$, so matrix A is defective for $\alpha = 0$.
- When $\alpha \neq= 0$, There are 2 linearly independent eigenvectors $(1, -2\alpha, 0)^T$ and $(0, 0, 1)^T$ corresponding to eigenvalues $\lambda_{1,2} = 0$. There are 3 linearly independent eigenvectors for matrix A. This shows that matrix A is not defective for $\alpha \neq= 0$.

So $\alpha = 0$ makes the matrix defective.

Problem 12

Let A be a diagonalizable $n \times n$ matrix. Prove that if B is any matrix that is similar to A, then B is diagonalizable.

Answer

A is diagonalizable, then $X^{-1}AX = D$. If B is similar to A then $A = S^{-1}BS$. Substitute A then we get $D = X^{-1}\left(S^{-1}BS\right)X = (SX)^{-1}B(SX)$. So, B is diagonalizable by SX.

Problem 13

Show that if A and B are two $n \times n$ matrices with the same diagonalizing matrix X, then $AB = BA$.

Answer

Assuming $A = XDX^{-1}$ and $B = XEX^{-1}$, where D, E are diagonal matrix and $DE = ED$. $AB = \left(XDX^{-1}\right)\left(XEX^{-1}\right) = XDEX^{-1} = XEDX^{-1} = \left(XEX^{-1}\right)\left(XDX^{-1}\right) = BA$.

Problem 14

Find the matrix associated with each of the following quadratic forms:
(a) $3x^2 - 5xy + y^2$
(b) $2x^2 + 3y^2 + z^2 + xy - 2xz + 3yz$

Answer

(a) $\begin{bmatrix} 3 & -\frac{5}{2} \\ -\frac{5}{2} & 1 \end{bmatrix}$

(b) $\begin{bmatrix} 2 & \frac{1}{2} & -1 \\ \frac{1}{2} & 3 & \frac{3}{2} \\ -1 & \frac{3}{2} & 1 \end{bmatrix}$

Problem 15

Show that if A is a symmetric positive definite matrix, then A is nonsingular and A^{-1} is also positive definite.

Answer

Let A be $n \times n$ symmetric positive definite matrix, we have that, $\det(A) > 0$, therefore A is nonsingular. Since $AA^{-1} = I_n$, we have that $\left(AA^{-1}\right)^T = \left(A^{-1}\right)^T A^T = (I_n)^T = I_n$. Therefore $\left(A^{-1}\right)^T = \left(A^T\right)^{-1}$. Since A is symmetric we have that $A = A^T$ and $\left(A^{-1}\right)^T = A^{-1}$ and therefore A^{-1} is symmetric. Since A is a positive definite matrix, all eigenvalues of A are positive and the eigenvalues of A^{-1} are the inverses of the eigenvalues of A. Hence since A^{-1} is symmetric we have that A is nonsingular and positive definite.